

OIL INDIA LIMITED — DRILLING DIRECTORATE

DULIAJAN, ASSAM · WELL COMPLETION REPORT (WCR) · SECTION 8

Well Name:	NAHARKATIYA-162 (NHK-162)	Field / Block:	Naharkatiya PML (NHK-II)
Spud Date:	14-JAN-2019	TD Reached:	19-JUN-2019 (3,620.0m MD)
Rig Assigned:	OIL-RIG-02 (2000 HP AC-SCR)	Status:	COMPLETED OIL PRODUCER
Target Horizon:	Tipam Sandstone TS-3 & Barail	Classification:	RESTRICTED SUBSURFACE ASSET

SECTION 4: GEOLOGICAL STRATIGRAPHY & FORMATION TOPS

Formation Name	Depth Top (m MD)	Depth Base (m MD)	Lithology & Characteristics
Alluvium / Dihing	0.0 m	310.0 m	Unconsolidated sands and pebble gravel beds
Dhekiajuli Formation	310.0 m	820.0 m	Loose porous sandstone, freshwater aquifer
Girujan Clay Formation	820.0 m	2,240.0 m	Reactive swelling bentonitic claystone
Tipam Sandstone (TS-1 to TS-6)	2,240.0 m	3,010.0 m	High permeability channel pay sands. Depleted pressure.
Barail Group	3,010.0 m	3,620.0 m	Overpressured coal-shale sequences and gas sands.

SECTION 8: HISTORICAL DRILLING HAZARDS & NPT LOG (PAGE 82)

INCIDENT: SEVERE LOST CIRCULATION AT 2,540m MD (TIPAM SANDSTONE TS-3)

Upon penetrating the micro-fractured porous Tipam Sandstone TS-3 interval at 2,540m MD, rig experienced sudden loss of mud returns with loss rate measuring 60 bbl/hr. Active suction pit volume dropped rapidly by 38 bbl. Standpipe pressure dropped from 2,600 psi to 1,850 psi due to loss of annular hydrostatic head.

FIELD-PROVEN MITIGATION & SOP: Immediately hoisted drillstring off-bottom to 2,500m to prevent differential sticking. Spotted 25 bbl coarse calcium carbonate (30 ppb) + medium mica (15 ppb) LCM pill across thief zone. Held 250 psi hesitation squeeze for 30 minutes. Reduced drilling mud weight from 11.2 ppg to 10.4 ppg. Full circulation restored at 14:00 hrs. Recommendation: In Tipam TS-3 intervals, maintain mud weight strictly within 10.2–10.6 ppg safe window. Keep 150 bags of coarse LCM staged on rig floor.

STATUTORY SIGN-OFF: Verified by B. Borah (Lead Mud Engineer, Oil India Ltd.) · OISD-STD-174 Compliant